

Question ID b96ff36e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: b96ff36e

In the xy -plane, the graph of the equation $(x - 3)^2 + (y - 5)^2 = 9$ is a circle. The point $(6, c)$, where c is a constant, lies on this circle. What is the value of c ?

ID: b96ff36e Answer

Correct Answer:

5

Rationale

The correct answer is **5**. It's given that in the xy -plane, the graph of the equation $(x - 3)^2 + (y - 5)^2 = 9$ is a circle. It's also given that the point $(6, c)$, where c is a constant, lies on this circle. It follows that the ordered pair $(6, c)$ makes the equation $(x - 3)^2 + (y - 5)^2 = 9$ true. Substituting **6** for x and c for y in this equation yields $(6 - 3)^2 + (c - 5)^2 = 9$, or $9 + (c - 5)^2 = 9$. Subtracting **9** from each side of this equation yields $(c - 5)^2 = 0$. It follows that the value of c is **5**.

Question Difficulty:

Medium

Question ID 82c8325f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 82c8325f

A circle in the xy -plane has its center at $(-4, 5)$ and the point $(-8, 8)$ lies on the circle. Which equation represents this circle?

- A. $(x - 4)^2 + (y + 5)^2 = 5$
- B. $(x + 4)^2 + (y - 5)^2 = 5$
- C. $(x - 4)^2 + (y + 5)^2 = 25$
- D. $(x + 4)^2 + (y - 5)^2 = 25$

ID: 82c8325f Answer

Correct Answer:

D

Rationale

Choice D is correct. A circle in the xy -plane can be represented by an equation of the form $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center of the circle and r is the length of a radius of the circle. It's given that the circle has its center at $(-4, 5)$. Therefore, $h = -4$ and $k = 5$. Substituting -4 for h and 5 for k in the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - (-4))^2 + (y - 5)^2 = r^2$, or $(x + 4)^2 + (y - 5)^2 = r^2$. It's also given that the point $(-8, 8)$ lies on the circle. Substituting -8 for x and 8 for y in the equation $(x + 4)^2 + (y - 5)^2 = r^2$ yields $(-8 + 4)^2 + (8 - 5)^2 = r^2$, or $(-4)^2 + (3)^2 = r^2$, which is equivalent to $16 + 9 = r^2$, or $25 = r^2$. Substituting 25 for r^2 in the equation $(x + 4)^2 + (y - 5)^2 = r^2$ yields $(x + 4)^2 + (y - 5)^2 = 25$. Thus, the equation $(x + 4)^2 + (y - 5)^2 = 25$ represents the circle.

Choice A is incorrect. The circle represented by this equation has its center at $(4, -5)$, not $(-4, 5)$, and the point $(-8, 8)$ doesn't lie on the circle.

Choice B is incorrect. The point $(-8, 8)$ doesn't lie on the circle represented by this equation.

Choice C is incorrect. The circle represented by this equation has its center at $(4, -5)$, not $(-4, 5)$, and the point $(-8, 8)$ doesn't lie on the circle.

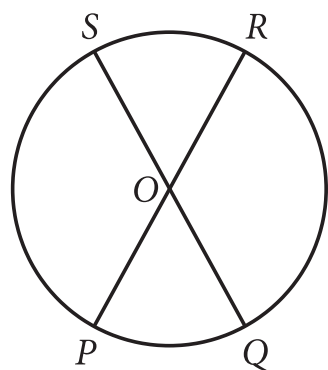
Question Difficulty:

Medium

Question ID 0815a5af

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 0815a5af



Note: Figure not drawn to scale.

The circle shown has center O , circumference 144π , and diameters $\overline{PR}$ and $\overline{QS}$. The length of arc PS is twice the length of arc PQ . What is the length of arc QR ?

- A. 24π
- B. 48π
- C. 72π
- D. 96π

ID: 0815a5af Answer

Correct Answer:
B

Rationale

Choice B is correct. Since $\overline{PR}$ and $\overline{QS}$ are diameters of the circle shown, $\overline{OS}$, $\overline{OR}$, $\overline{OP}$, and $\overline{OQ}$ are radii of the circle and are therefore congruent. Since $\angle SOP$ and $\angle ROQ$ are vertical angles, they are congruent. Therefore, arc PS and arc QR are formed by congruent radii and have the same angle measure, so they are congruent arcs. Similarly, $\angle SOR$ and $\angle POQ$ are vertical angles, so they are congruent. Therefore, arc SR and arc PQ are formed by congruent radii and have the same angle measure, so they are congruent arcs. Let x represent the length of arc SR . Since arc SR and arc PQ are congruent arcs, the length of arc PQ can also be represented by x . It's given that the length of arc PS is twice the length of arc PQ . Therefore, the length of arc PS can be represented by the expression $2x$. Since arc PS and arc QR are congruent arcs, the length of arc QR can also be represented by $2x$. This gives the expression $x + x + 2x + 2x$. Since it's given that the circumference is 144π , the expression $x + x + 2x + 2x$ is equal to 144π . Thus $x + x + 2x + 2x = 144\pi$, or $6x = 144\pi$. Dividing both sides of this equation by 6 yields $x = 24\pi$. Therefore, the length of arc QR is $2(24\pi)$, or 48π .

Choice A is incorrect. This is the length of arc PQ , not arc QR .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

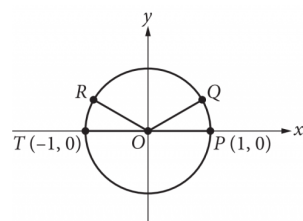
Question Difficulty:

Medium

Question ID 95ba2d09

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 95ba2d09



In the xy -plane above, points P , Q , R , and T lie on the circle with center O . The degree measures of angles POQ and ROT are each 30° . What is the radian measure of angle QOR ?

- A. $\frac{5}{6} \pi$
- B. $\frac{3}{4} \pi$
- C. $\frac{2}{3} \pi$
- D. $\frac{1}{3} \pi$

ID: 95ba2d09 Answer

Correct Answer:
C

Rationale

Choice C is correct. Because points T , O , and P all lie on the x -axis, they form a line. Since the angles on a line add up to 180° , and it's given that angles POQ and ROT each measure 30° , it follows that the measure of angle QOR is $180^\circ - 30^\circ - 30^\circ = 120^\circ$. Since the arc of a complete circle is 360° or 2π radians, a proportion can be set up to convert the measure of angle QOR from degrees to radians: $\frac{360 \text{ degrees}}{2 \pi \text{ radians}} = \frac{120 \text{ degrees}}{x \text{ radians}}$, where x is the radian measure of angle QOR . Multiplying each side of the proportion by $2 \pi x$ gives $360x = 240 \pi$. Solving for x gives $\frac{240}{360} \pi$, or $\frac{2}{3} \pi$.

Choice A is incorrect and may result from subtracting only angle POQ from 180° to get a value of 150° and then finding the radian measure equivalent to that value. Choice B is incorrect and may result from a calculation error. Choice D is incorrect and may result from calculating the sum of the angle measures, in radians, of angles POQ and ROT .

Question Difficulty:
Medium

Question ID 856372ca

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 856372ca

In the xy -plane, a circle with radius 5 has center $(-8,6)$. Which of the following is an equation of the circle?

- A. $(x-8)^2+(y+6)^2=25$
- B. $(x+8)^2+(y-6)^2=25$
- C. $(x-8)^2+(y+6)^2=5$
- D. $(x+8)^2+(y-6)^2=5$

ID: 856372ca Answer

Correct Answer:
B

Rationale

Choice B is correct. An equation of a circle is $(x-h)^2+(y-k)^2=r^2$, where the center of the circle is (h,k) and the radius is r . It's given that the center of this circle is $(-8,6)$ and the radius is 5. Substituting these values into the equation gives $(x-(-8))^2+(y-6)^2=5^2$, or $(x+8)^2+(y-6)^2=25$.

Choice A is incorrect. This is an equation of a circle that has center $(8,-6)$. Choice C is incorrect. This is an equation of a circle that has center $(8,-6)$ and radius $\sqrt{5}$. Choice D is incorrect. This is an equation of a circle that has radius $\sqrt{5}$.

Question Difficulty:
Medium

Question ID 2d521ca9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 2d521ca9

The measure of angle Z is 60° . What is the measure, in radians, of angle Z ?

- A. $\frac{1}{6}\pi$
- B. $\frac{1}{3}\pi$
- C. $\frac{2}{3}\pi$
- D. 1π

ID: 2d521ca9 Answer

Correct Answer:

B

Rationale

Choice B is correct. The measure of an angle, in radians, can be found by multiplying its measure, in degrees, by $\frac{\pi}{180}$. It's given that the measure of angle Z is 60° . It follows that the measure, in radians, of angle Z is $60\left(\frac{\pi}{180}\right)$, or $\frac{1}{3}\pi$.

Choice A is incorrect. This is the measure, in radians, of an angle whose measure is 30° , not 60° .

Choice C is incorrect. This is the measure, in radians, of an angle whose measure is 120° , not 60° .

Choice D is incorrect. This is the measure, in radians, of an angle whose measure is 180° , not 60° .

Question Difficulty:

Medium

Question ID f1c1e971

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: f1c1e971

The measure of angle R is $\frac{2\pi}{3}$ radians. The measure of angle T is $\frac{5\pi}{12}$ radians greater than the measure of angle R . What is the measure of angle T , in degrees?

- A. 75
- B. 120
- C. 195
- D. 390

ID: f1c1e971 Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that the measure of angle R is $\frac{2\pi}{3}$ radians, and the measure of angle T is $\frac{5\pi}{12}$ radians greater than the measure of angle R . Therefore, the measure of angle T is equal to $\frac{2\pi}{3} + \frac{5\pi}{12}$ radians. Multiplying $\frac{2\pi}{3}$ by $\frac{4}{4}$ to get a common denominator with $\frac{5\pi}{12}$ yields $\frac{8\pi}{12}$. Therefore, $\frac{2\pi}{3} + \frac{5\pi}{12}$ is equivalent to $\frac{8\pi}{12} + \frac{5\pi}{12}$, or $\frac{13\pi}{12}$. Therefore, the measure of angle T is $\frac{13\pi}{12}$ radians. The measure of angle T , in degrees, can be found by multiplying its measure, in radians, by $\frac{180}{\pi}$. This yields $\frac{13\pi}{12} \times \frac{180}{\pi}$, which is equivalent to **195** degrees. Therefore, the measure of angle T is **195** degrees.

Choice A is incorrect. This is the number of degrees that the measure of angle T is greater than the measure of angle R .

Choice B is incorrect. This is the measure of angle R in degrees.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID a0cacec1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: a0cacec1

An angle has a measure of $\frac{16\pi}{15}$ radians. What is the measure of the angle, in degrees?

ID: a0cacec1 Answer

Correct Answer:
192

Rationale

The correct answer is **192**. The measure of an angle, in degrees, can be found by multiplying its measure, in radians, by $\frac{180 \text{ degrees}}{\pi \text{ radians}}$. Multiplying the given angle measure, $\frac{16\pi}{15}$ **radians**, by $\frac{180 \text{ degrees}}{\pi \text{ radians}}$ yields $\left(\frac{16\pi}{15} \text{ radians}\right) \left(\frac{180 \text{ degrees}}{\pi \text{ radians}}\right)$, which simplifies to **192** degrees.

Question Difficulty:
Medium

Question ID ee540927

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: ee540927

$x^2 + 58x + y^2 = 0$

In the xy -plane, the graph of the given equation is a circle. What are the coordinates (x, y) of the center of the circle?

- A. $(0, 29)$
- B. $(0, -29)$
- C. $(29, 0)$
- D. $(-29, 0)$

ID: ee540927 Answer

Correct Answer:
D

Rationale

Choice D is correct. It’s given that in the xy -plane, the graph of $x^2 + 58x + y^2 = 0$ is a circle. The equation of a circle in the xy -plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the coordinates of the center of the circle are (h, k) and the radius of the circle is r . By completing the square, the equation $x^2 + 58x + y^2 = 0$ can be rewritten as $\left(x^2 + 58x + \left(\frac{58}{2}\right)^2\right) + y^2 = 0 + \left(\frac{58}{2}\right)^2$, or $(x^2 + 58x + 841) + y^2 = 841$. This equation is equivalent to $(x + 29)^2 + y^2 = 841$, or $(x - (-29))^2 + (y - 0)^2 = 841$. Therefore, h is -29 and k is 0 , and the coordinates (x, y) of the center of the circle are $(-29, 0)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID 74d8b897

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 74d8b897

An angle has a measure of $\frac{9\pi}{20}$ radians. What is the measure of the angle in degrees?

ID: 74d8b897 Answer

Correct Answer:

81

Rationale

The correct answer is **81**. The measure of an angle, in degrees, can be found by multiplying its measure, in radians, by $\frac{180 \text{ degrees}}{\pi \text{ radians}}$. Multiplying the given angle measure, $\frac{9\pi}{20}$ radians, by $\frac{180 \text{ degrees}}{\pi \text{ radians}}$ yields $\left(\frac{9\pi}{20} \text{ radians}\right)\left(\frac{180 \text{ degrees}}{\pi \text{ radians}}\right)$, which is equivalent to **81** degrees.

Question Difficulty:

Medium

Question ID fc8aa563

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: fc8aa563

What is the center of the circle in the xy -plane defined by the equation $(x - 1)^2 + (y + 7)^2 = 1$?

- A. $(-1, -7)$
- B. $(-1, 7)$
- C. $(1, -7)$
- D. $(1, 7)$

ID: fc8aa563 Answer

Correct Answer:
C

Rationale

Choice C is correct. The equation of a circle in the xy -plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r . It's given that the circle in the xy -plane is defined by the equation $(x - 1)^2 + (y + 7)^2 = 1$. This equation can be written as $(x - 1)^2 + (y - (-7))^2 = 1$. For this equation, it follows that $h = 1$ and $k = -7$. Therefore, the center of the circle in the xy -plane defined by the given equation is $(1, -7)$.

Choice A is incorrect. This is the center of the circle in the xy -plane that is defined by the equation $(x + 1)^2 + (y + 7)^2 = 1$, not $(x - 1)^2 + (y + 7)^2 = 1$.

Choice B is incorrect. This is the center of the circle in the xy -plane that is defined by the equation $(x + 1)^2 + (y - 7)^2 = 1$, not $(x - 1)^2 + (y + 7)^2 = 1$.

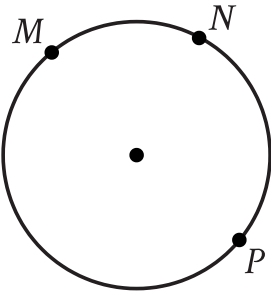
Choice D is incorrect. This is the center of the circle in the xy -plane that is defined by the equation $(x - 1)^2 + (y - 7)^2 = 1$, not $(x - 1)^2 + (y + 7)^2 = 1$.

Question Difficulty:
Medium

Question ID 800e71b8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 800e71b8



Points M , N , and P lie on the circle shown. On this circle, minor arc MN has a length of **39** centimeters and major arc MPN has a length of **195** centimeters. What is the circumference, in centimeters, of the circle shown?

- A. **39**
- B. **156**
- C. **195**
- D. **234**

ID: 800e71b8 Answer

Correct Answer:
D

Rationale

Choice D is correct. Since the endpoints of minor arc MN and major arc MPN are the same, and the arcs together form a full circle, the sum of the lengths of these two arcs is equal to the circumference of the circle. It's given that the length of minor arc MN is **39** centimeters and the length of major arc MPN is **195** centimeters. Therefore, the circumference of the circle, in centimeters, is **39 + 195**, or **234**.

Choice A is incorrect. This is the length, in centimeters, of minor arc MN , not the circumference, in centimeters, of the circle.

Choice B is incorrect. This is the difference of the lengths of major arc MPN and minor arc MN , in centimeters.

Choice C is incorrect. This is the length, in centimeters, of major arc MPN , not the circumference, in centimeters, of the circle.

Question Difficulty:
Medium

Question ID 8e7689e0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 8e7689e0

The number of radians in a 720-degree angle can be written as $a\pi$, where a is a constant. What is the value of a ?

ID: 8e7689e0 Answer

Rationale

The correct answer is 4. There are π radians in a 180° angle. An angle measure of 720° is 4 times greater than an angle measure of 180° . Therefore, the number of radians in a 720° angle is 4π .

Question Difficulty:
Medium

Question ID aa2d36fe

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: aa2d36fe

A circle in the xy -plane has the equation $(x - 13)^2 + (y - k)^2 = 64$. Which of the following gives the center of the circle and its radius?

- A. The center is at $(13, k)$ and the radius is 8.
- B. The center is at $(k, 13)$ and the radius is 8.
- C. The center is at $(k, 13)$ and the radius is 64.
- D. The center is at $(13, k)$ and the radius is 64.

ID: aa2d36fe Answer

Correct Answer:

A

Rationale

Choice A is correct. For a circle in the xy -plane that has the equation $(x - h)^2 + (y - k)^2 = r^2$, where h , k , and r are constants, (h, k) is the center of the circle and the positive value of r is the radius of the circle. In the given equation, $h = 13$ and $r^2 = 64$. Taking the square root of each side of $r^2 = 64$ yields $r = \pm 8$. Therefore, the center of the circle is at $(13, k)$ and the radius is 8.

Choice B is incorrect. This gives the center and radius of a circle with equation $(x - k)^2 + (y - 13)^2 = 64$, not $(x - 13)^2 + (y - k)^2 = 64$.

Choice C is incorrect. This gives the center and radius of a circle with equation $(x - k)^2 + (y - 13)^2 = 4,096$, not $(x - 13)^2 + (y - k)^2 = 64$.

Choice D is incorrect. This gives the center and radius of a circle with equation $(x - 13)^2 + (y - k)^2 = 4,096$, not $(x - 13)^2 + (y - k)^2 = 64$.

Question Difficulty:

Medium